

- [Charts: visualizing data](#)
- [Getting started with BigQuery](#)

Machine Learning

These are a few of the notebooks related to Machine Learning, including Google's online Machine Learning course. See the [full course website](#) for more.

- [Intro to Pandas DataFrame](#)
- [Intro to RAPIDS cuDF to accelerate pandas](#)
- [Getting Started with cuML's accelerator mode](#)

Using Accelerated Hardware

- [Train a CNN to classify handwritten digits on the MNIST dataset using Flax NNX API](#)
- [Train a Vision Transformer \(ViT\) for image classification with JAX](#)
- [Text classification with a transformer language model using JAX](#)

✓ Featured examples

- [Train a miniGPT language model with JAX AI Stack](#)
- [LoRA/QLoRA finetuning for LLM using Tunix](#)
- [Parameter-efficient fine-tuning of Gemma with LoRA and QLoRA](#)
- [Loading Hugging Face Transformers Checkpoints](#)
- [8-bit Integer Quantization in Keras](#)
- [Float8 training and inference with a simple Transformer model](#)
- [Pretraining a Transformer from scratch with KerasHub](#)
- [Simple MNIST convnet](#)
- [Image classification from scratch using Keras 3](#)
- [Image Classification with KerasHub](#)

```
import numpy as np
import matplotlib.pyplot as plt

sales_data = np.array([50000, 60000, 70000, 85000])

quarters = ["Q1", "Q2", "Q3", "Q4"]

total_sales = np.sum(sales_data)

percentage_increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100

print("Total Sales for the Year:", total_sales)
print("Percentage Increase from Q1 to Q4: {:.2f}%".format(percentage_increase))

plt.figure(figsize=(8,5))

plt.plot(quarters, sales_data, marker="o")

for i, value in enumerate(sales_data):
    plt.text(i, value + 1000, str(value), ha="center")

plt.title("Quarterly Sales Performance")
plt.xlabel("Quarter")
plt.ylabel("Sales")
plt.grid(True)

plt.show()
```

Total Sales for the Year: 265000
Percentage Increase from Q1 to Q4: 70.00%

